

Metalens Technology

A Deep Dive into Flat Optics and Nanoscale Phase Engineering

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OPTICS

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METASURFACES

What is a Metalens?

A metalens is a **flat lens** comprising a metasurface or nanostructure with circular symmetry, precisely engineered to generate a radial phase profile $\phi(r)$.

A **metasurface** is an array of nanoscale optical elements — also known as **meta-atoms, nano-scatterers, nano-elements, nano-antennas, or nanobricks** — arranged on a flat surface. By properly designing these nano-elements, a metasurface can **transform an incident plane wavefront into a desired wavefront**, such as a spherical wavefront for focusing.

Wavelength-Scale Interaction

The size and spacing of meta-atoms are typically **comparable to or smaller than the wavelength of light**, enabling unique light-matter interactions.

Local Phase Control

Unlike conventional optics, where phase accumulates through **propagation inside a material**, a metasurface introduces a desired **phase shift locally at the surface itself**.

The Law of Refraction and Reflection for a Metasurface

A metasurface can introduce a **spatially varying phase profile** across its surface, creating a phase gradient $\nabla\Phi$. Because of this spatial variation, the conventional laws of reflection and refraction are modified.

Generalized Snell's Law

$$n_t \sin \theta_t - n_i \sin \theta_i = \frac{\lambda_0}{2\pi} \frac{d\Phi}{dx}$$

The phase gradient term modifies the conventional relationship between incident and transmitted angles.

Classical Snell's Law (Special Case)

$$n_i \sin \theta_i = n_t \sin \theta_t$$

When the phase profile is constant, $\frac{d\Phi}{dx} = 0$, and the generalized law reduces to the well-known Snell's law.

- A metasurface controls the direction of light by introducing a **spatial phase gradient** — a capability unavailable in conventional flat interfaces.

The generalized Snell's law reveals that the deflection angle of transmitted light depends directly on the local slope of the phase profile. By engineering $\frac{d\Phi}{dx}$ at every point, arbitrary beam steering and wavefront shaping become possible on a flat substrate.

How Does Phase Shift Correspond to Time Retardation?

A phase shift can be understood as a **time delay or time retardation** of a wave. For a monochromatic wave:

$$\Phi = \omega \Delta t$$

Φ

Phase shift

ω

Angular frequency

Δt

Time retardation

Physical Interpretation

When different meta-atoms introduce different phase shifts, they effectively introduce **different time retardations across the surface**. Although the light does not necessarily travel through different physical distances, different points on the metasurface make the wave **advance or lag in phase** relative to one another.

This spatial variation in phase is what allows a flat metasurface to **reshape the wavefront** — for example, transforming a plane wave into a spherical wavefront and focusing it to a point.

Derivation of the Refraction Law from Huygens's Principle

According to Huygens's principle, a plane wavefront propagates as if composed of many point sources, each emitting a spherical wave, such that the wavefront at a later time is the envelope of these wavelets. A metasurface introduces different phase shifts at every point – equivalent to introducing different time retardations across the surface.

Setup

Consider a metasurface between two media with refractive indices n_1 and n_2 . For the two parts of the wavefront to reach the new wavefront at the same time:

$$t_{AD} + \tau_A = t_{BC} + \tau_C$$

From the geometry:

$$t_{AD} = \frac{n_2}{c} \Delta r \sin \theta_2, \quad t_{BC} = \frac{n_1}{c} \Delta r \sin \theta_1$$

The phase shift is related to the time retardation by $\Phi = \omega \tau$. Using $k_0 = \omega/c$:

$$n_2 \sin \theta_2 - n_1 \sin \theta_1 = \frac{1}{k_0} \frac{\Delta \Phi}{\Delta r}$$

Result

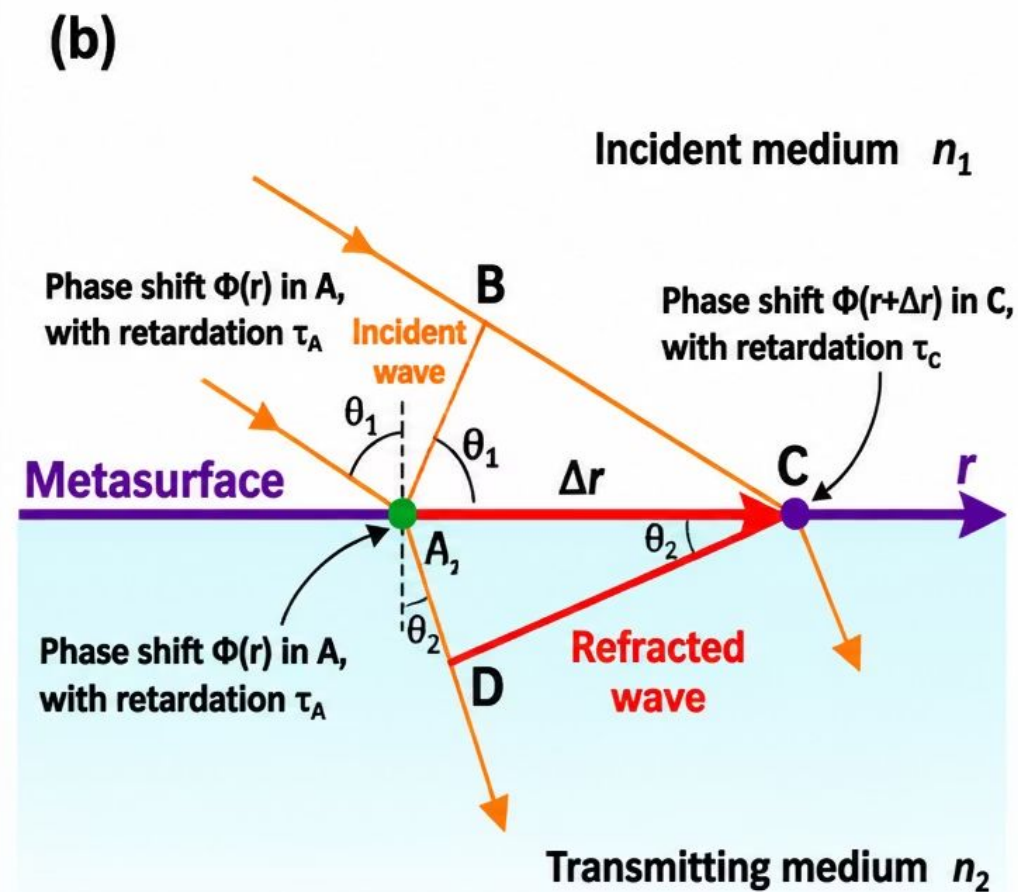
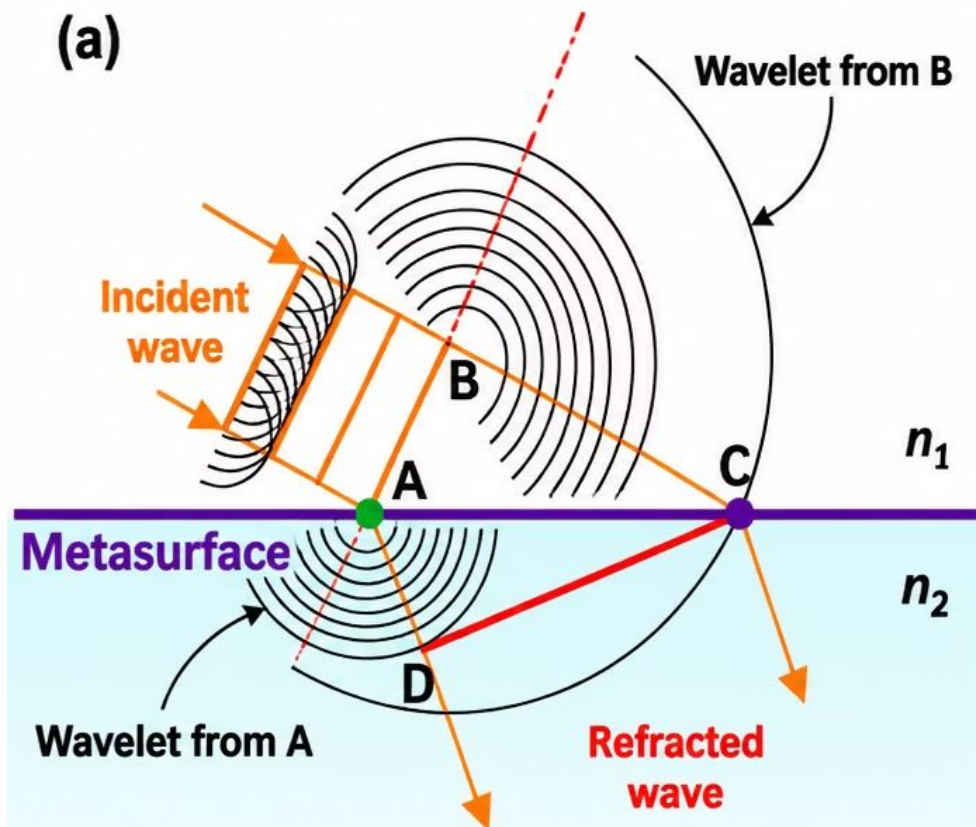
Taking the limit $\Delta r \rightarrow 0$:

$$n_2 \sin \theta_2 - n_1 \sin \theta_1 = \frac{1}{k_0} \frac{d\Phi}{dr}$$

Equivalently:

$$n_2 \sin \theta_2 - n_1 \sin \theta_1 = \frac{\lambda_0}{2\pi} \frac{d\Phi}{dr}$$

❏ **A spatial phase gradient introduced by the metasurface modifies the conventional Snell's law.**



Metalens Phase Profile

Different Types of Phase Profiles

Metalenses can exhibit various types of phase profiles tailored to specific applications. These profiles include radial, azimuthal, and even complex multifocal patterns.



Radial Hyperbolic

Concentrates light towards a central point, akin to traditional lenses. The phase increases with radial distance to equalize optical path lengths to the focal point.



Azimuthal / Spiral

The sum of radial and spiral phases. These profiles may perform **edge-enhanced imaging**, enabling applications beyond simple focusing.



Multifocal

Allows for **simultaneous focusing at multiple points**, enabling depth-of-field extension and multi-plane imaging in a single flat element.

Metalens Phase Profile: Radial Hyperbolic

For a metalens, a **radial hyperbolic phase profile** is required. The derivation proceeds from the requirement that all rays converge to the focal point F .

1. Optical Path Difference

Consider a point on the metalens at radial distance r from its center. The distance from this point to the focal point is $\sqrt{r^2 + f^2}$, while the distance from the center to the focus is simply f . The additional optical path that the ray at r must travel is:

$$\ell_r = \sqrt{r^2 + f^2} - f$$

2. Convert Optical Path into Phase

An optical path difference OPD corresponds to a phase shift $\Phi = k \cdot \text{OPD}$, where $k = \frac{2\pi}{\lambda}$. Therefore, the required phase shift at radial position r is:

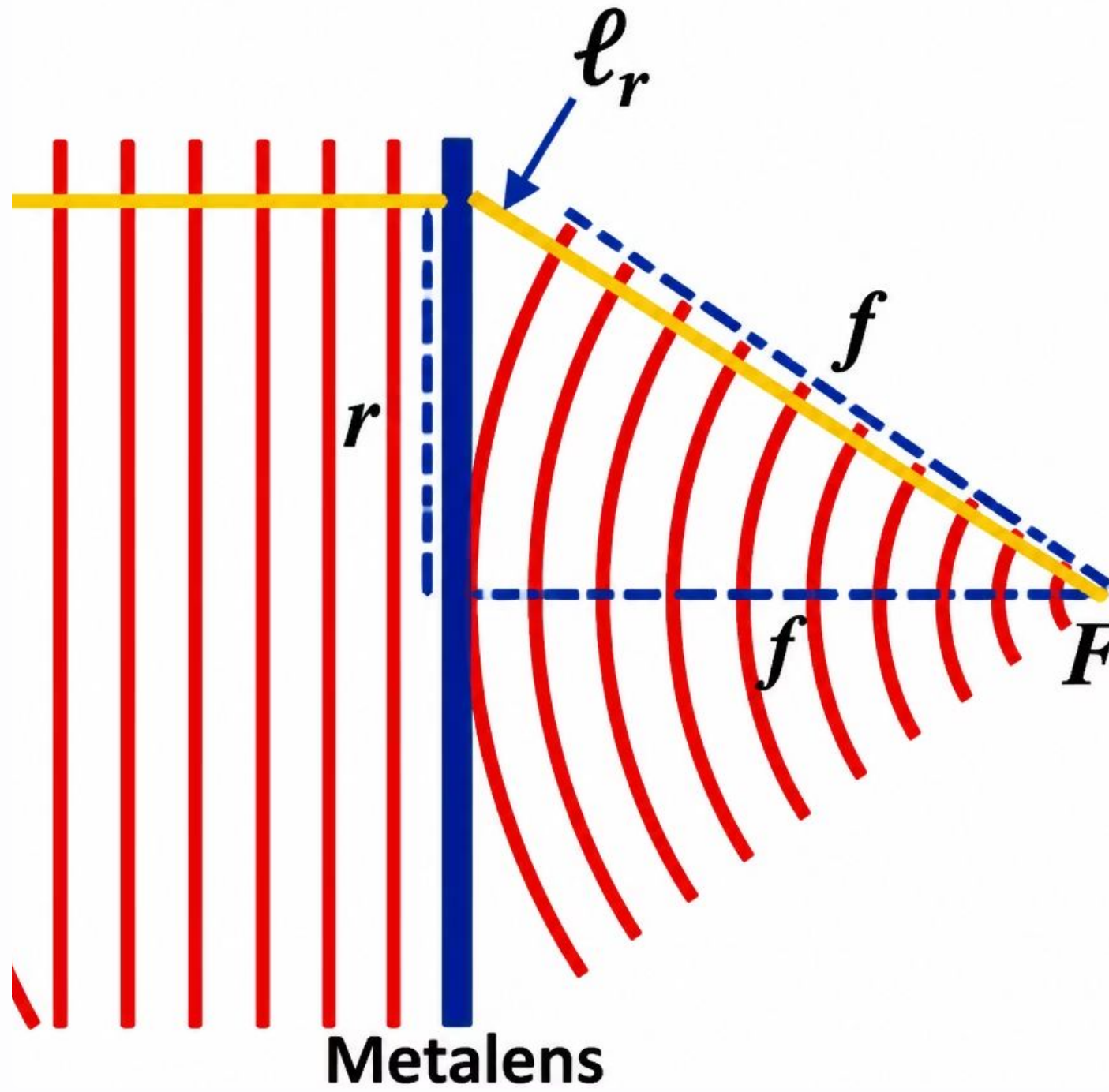
$$\Phi(r) = \pm k \left(\sqrt{r^2 + f^2} - f \right)$$

This is the **hyperbolic phase profile** of a metalens.

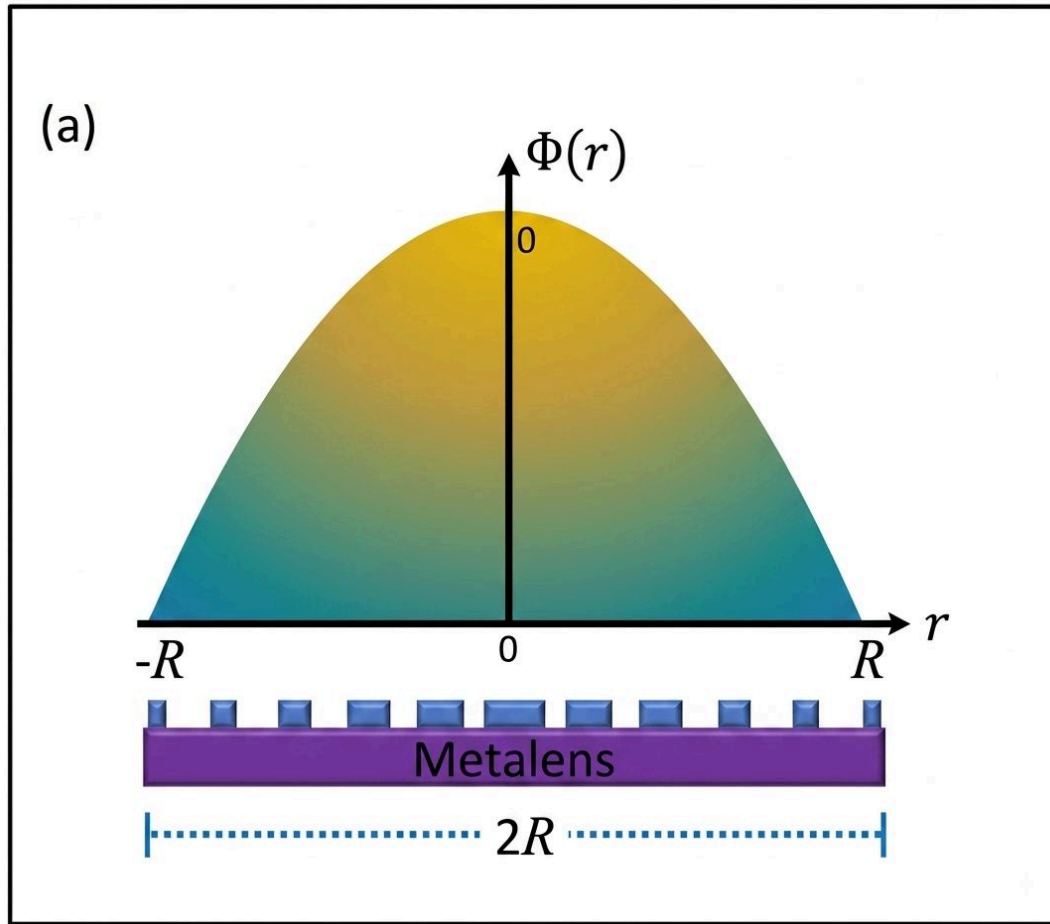
Physical Meaning

□ **The phase shift increases with radial distance r so that all rays, despite traveling different distances, arrive at the focal point with the correct phase.**

At the center ($r = 0$), no additional phase is needed. As r increases, the ray must travel a longer path to reach the focus, so the metasurface must compensate by introducing a larger phase shift. The hyperbolic form ensures exact phase matching across the entire aperture.



Gradient of Phase Profile



Shows the phase profile in a metalens of radius R , which determines its focusing properties and the focal distance f .

The gradient of the phase profile $\frac{d\Phi}{dr}$ determines the local deflection angle of transmitted light at each radial position. Near the center, the gradient is small and rays are deflected only slightly. At the edge of the aperture, the gradient is large, bending rays steeply toward the focal point.

- The hyperbolic profile ensures that the gradient at every point satisfies the generalized Snell's law, directing all rays to converge at f .

From Phase Profile to Metasurface Profile

The metalens phase profile $\Phi(r)$ can be reproduced by radially varying the geometrical parameters of nanoelements – such as size, shape, or orientation – across the flat surface.

Design Complexity

The phase shift introduced by meta-atoms depends on many parameters: refractive index, height, absorption, unit cell size, and wavelength.

Although some semi-analytical models have been reported, there is **no single mathematical relation or analytic deduction** between Φ and the size or shape of meta-atoms.

Such a relationship is obtained by **numerical calculations** with:

FDTD

Finite Difference Time Domain

RCWA

Rigorous Coupled Wave
Analysis

Empirical Phase Model

For example, the phase shift introduced by dielectric nanocylinders or nanoposts may be approximated by:

$$\Phi = K_1 D^2 + K_2 D + K_3$$

where D is the post diameter, and $K_1 - K_3$ are constants that may depend on several physical and geometrical parameters such as meta-atom refractive index and height.

- ❏ This empirical quadratic relationship is geometry-specific and must be re-derived numerically for each new material system or wavelength regime.

Meta-Atoms

A metalens is usually composed of **dielectric meta-atoms** designed to introduce a desired phase shift to the transmitted light. These meta-atoms are typically made from **high-refractive-index, low-loss dielectric materials**. A high refractive index strongly **confines light** inside the nanoelement, allows a larger **phase shift** to be introduced, and reduces the required height to achieve full 2π phase coverage.

Three Mechanisms for Phase Control

1 Geometrical Phase

Based on the **Pancharatnam–Berry (PB) phase**. The phase is controlled by the **orientation angle** of asymmetric meta-atoms such as nanofins. Generally requires **circularly polarized light**.

2 Propagation Phase

The phase is accumulated as light **propagates through the meta-atom**. Controlled by changing the **height and effective refractive index** of the nanoelement.

3 Resonant Phase

The phase is controlled by tuning the **optical resonances** of the meta-atom, closely related to **Mie-type resonances** in high-index dielectric nanoelements.

From Ideal to Real Metalenses

Ideal Assumption

Every point provides exactly the required phase – a continuous, perfect phase profile.

Practical Limitation

Meta-atoms have a **finite size** and must be placed with finite spacing, causing the real phase profile to deviate from the ideal target.

Design Goal

Optimize meta-atom geometries for **aberration correction, new functionalities, and application-specific designs** so that the real phase profile approaches the desired profile as closely as possible.